

AERIAL ROBOTICS CHALLENGES

AZMI BIN BASHARUDIN

2211387

Future Capabilities



- Full Autonomy: Beyond programmed tasks—understanding intent and making decisions.
- Self-Healing: Robots detecting and repairing own faults.
- Cross-Domain Operations: Air-ground-sea systems working together.
- Human-Robot Collaboration: Safe, intelligent cooperation with humans.



Ethical Considerations



- Surveillance & Privacy: Drone surveillance, facial recognition.
- Lethal Autonomous Weapons (LAWs): Debate around military applications.
- Job Displacement: Replacing human workers in manual/low-skill tasks.



Sustainability Issues



- Power Consumption: High for long-duration or high-load robots.
- Battery Waste: Toxic elements in lithium batteries.
- Material Recycling: Components like rare earth magnets and PCBs are hard to recycle.
- Green Robotics: Use of biodegradable materials, solar charging, and sustainable design.



Thank You!

